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(54) **SMOKER FOR GRILL**

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**Related U.S. Application Data**

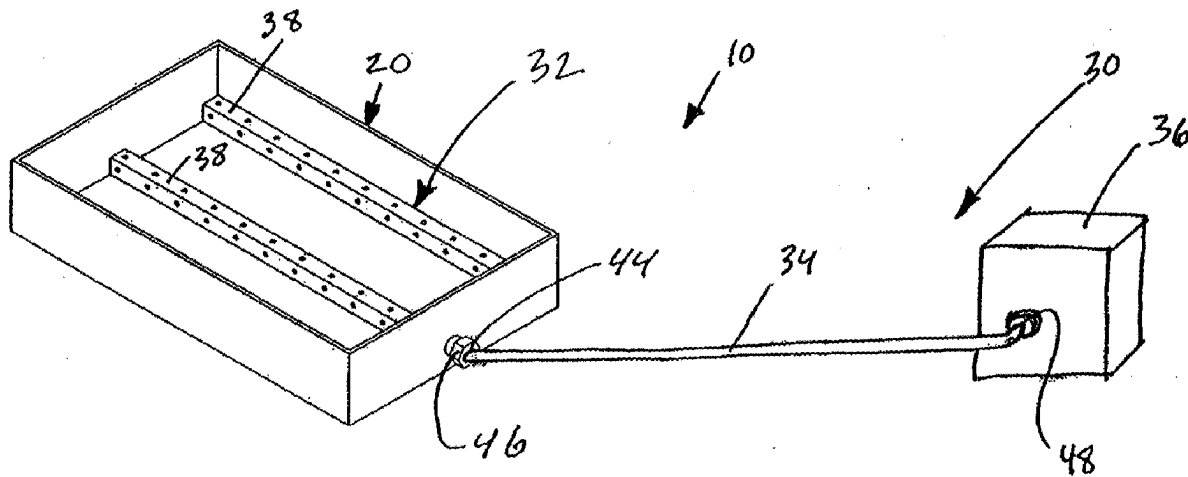
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(57)

**ABSTRACT**

A smoker for use with a grill, comprising a smoke-releasing material holder positionable between a heat source and food rack within a grill, at least one air diffuser positioned at least partially within the holder in air transmitting communication with the holder, and an air actuator coupled to the at least one air diffuser in air transmitting communication with the at least one air diffuser.



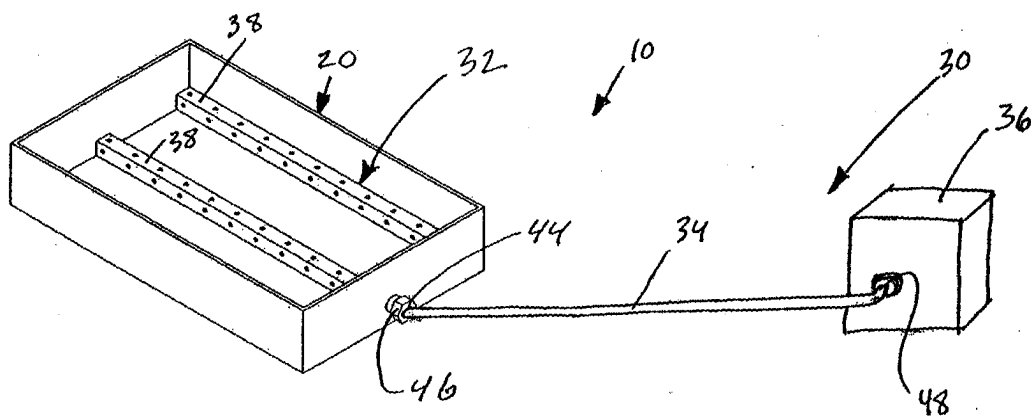


FIG. 1

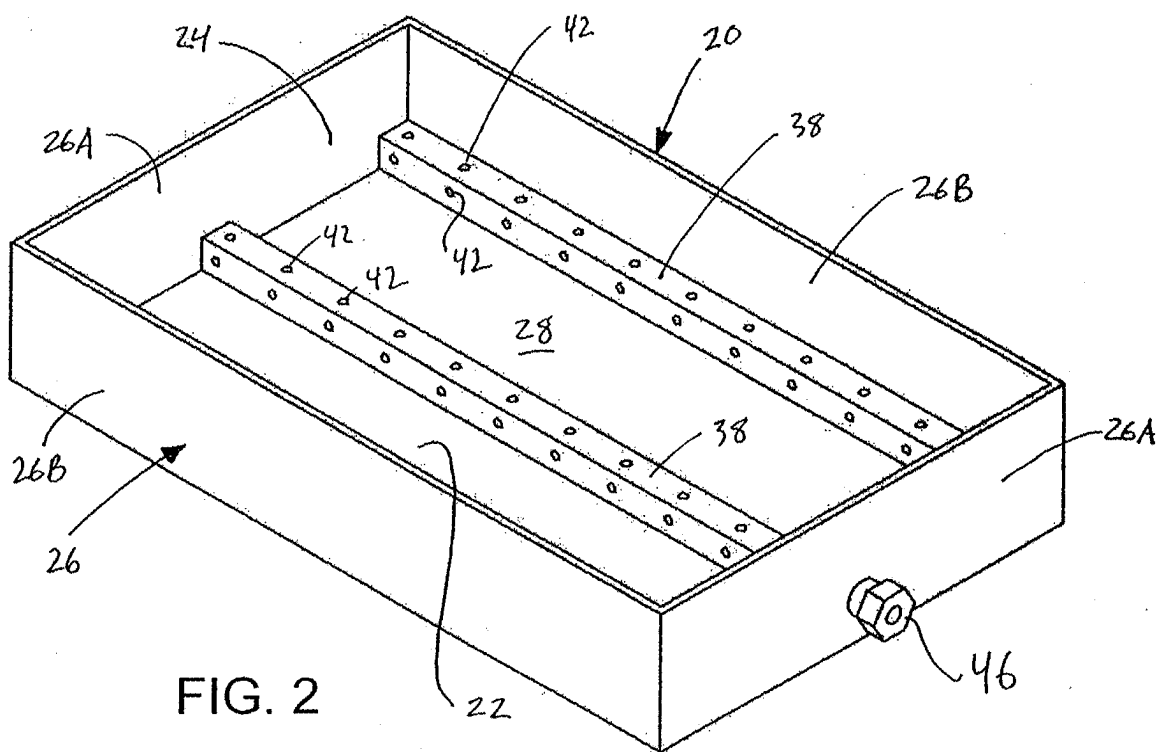
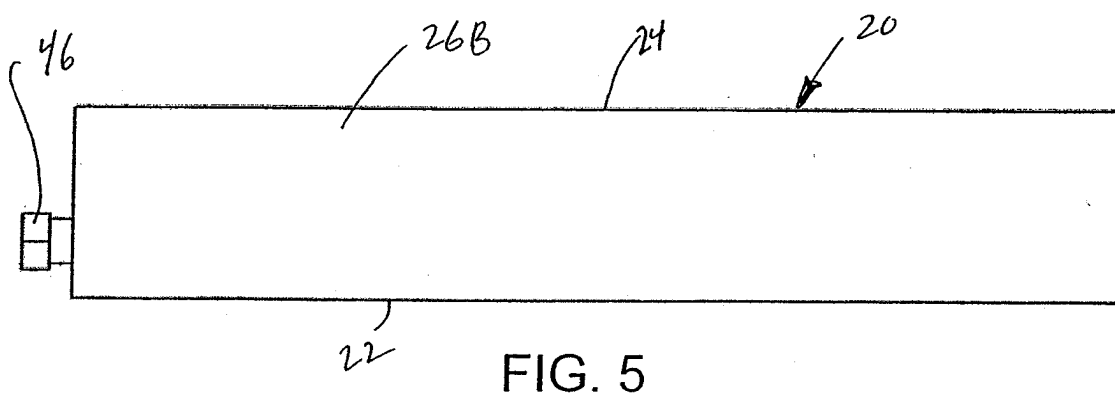
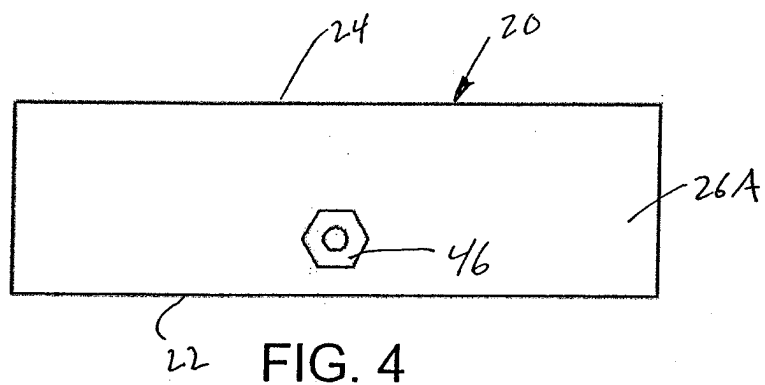
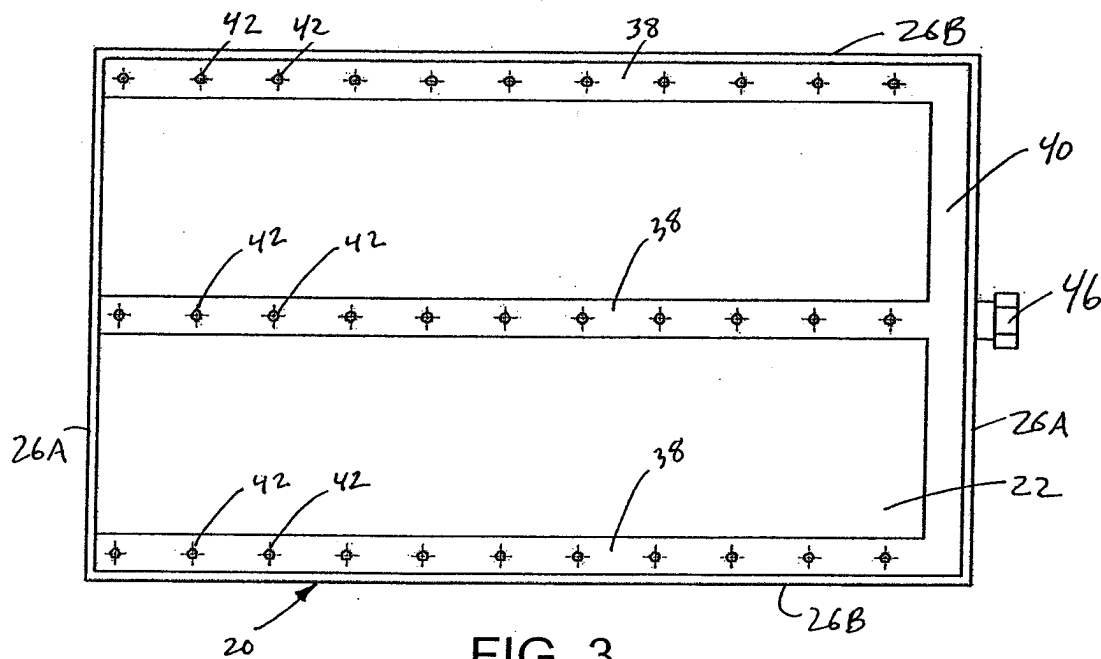


FIG. 2



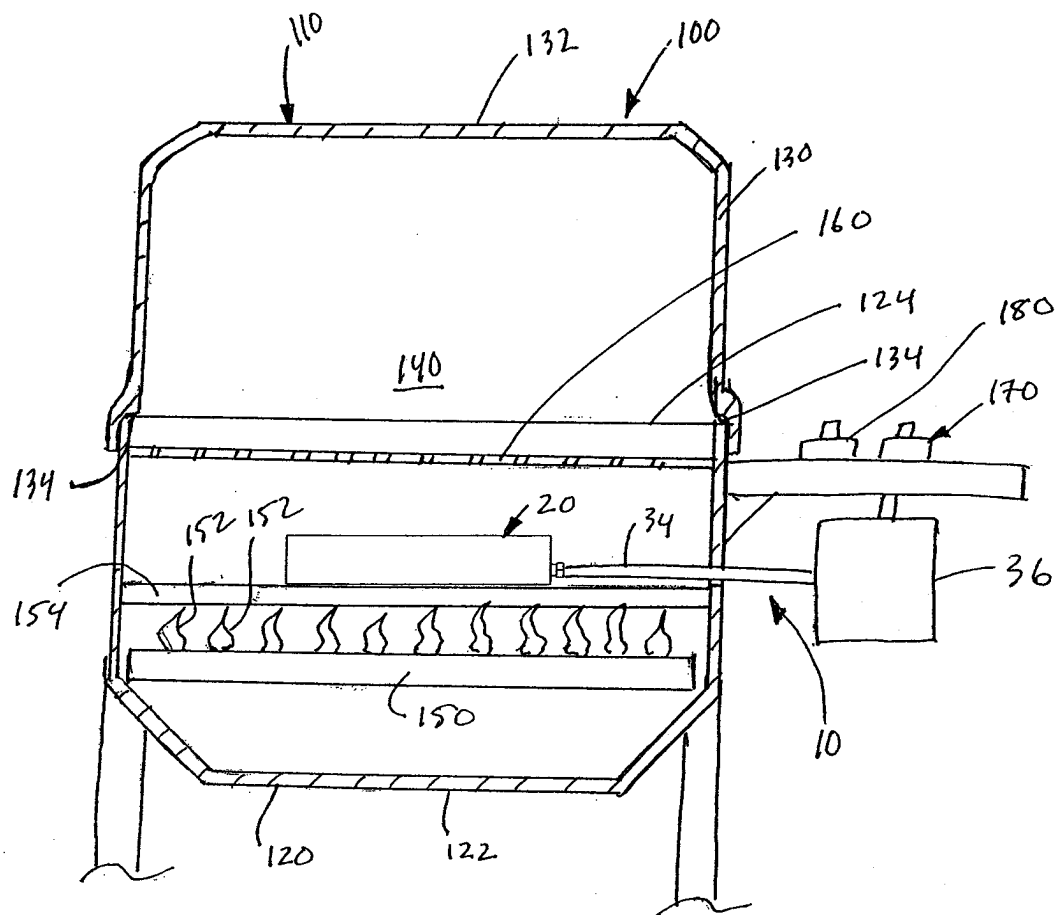


FIG. 6

## SMOKER FOR GRILL

### PRIORITY CLAIM

**[0001]** This is a continuation of U.S. application Ser. No. 14/484,156, filed Sep. 11, 2014, which is divisional of U.S. application Ser. No. 12/157,778, filed on Jun. 12, 2008, which claims the benefit of U.S. Provisional Application No. 60/937,373, filed Jun. 26, 2007, all of which are incorporated herein in their entirety.

### FIELD

**[0002]** The present invention relates to food cooking apparatus, and more particularly, to a smoker for food-cooking grills.

### BACKGROUND

**[0003]** Using smoke to flavor food is widely known in the art. For example, stand-alone smoker devices have been developed solely to generate smoke for flavoring food. Typically, a stand-alone smoker is in smoke transmitting communication with a food cooking apparatus, such as a grill, to introduce smoke into a food cooking environment defined by the food cooking apparatus. Food in the food cooking environment is cooked by a heat source of the food cooking apparatus and the smoke from the stand-alone smoker flavors the food.

**[0004]** While generally efficient at producing smoke, stand-alone smokers can be bulky, expensive and impractical. In order to smoke food using a stand-alone smoker, a grill-owner must have adequate space to store and operate the smoker, which, owing to the size of conventional stand-alone smokers, may be unavailable. Further, many grill owners only occasionally desire to smoke food. Also, maintaining a stand-alone, dedicated smoker may be time consuming and costly. Accordingly, an expensive, dedicated smoker may be economically and physically impractical for such intermittent use.

**[0005]** In other conventional systems, a smoke-generating device is positioned within a food cooking environment to create a combination cooker/smoker system. For example, in some systems, an outdoor grill defines a food cooking environment that has a heat source, such as a gas burner or charcoal, positioned at a lower portion of the environment, a food rack positioned above the heat source, and a smoke-generating device positioned intermediate the heat source and the food rack. Typical smoke-generating devices include fixed or removable trays containing smoke-releasing material, such as wood chips or pellets. Heat from the heat source causes the smoke-releasing material to ignite or smolder, which results in smoke being released from the material. The generated smoke circulates through the food cooking environment to flavor the food on the rack.

**[0006]** Conventional cooker/smoker systems, such as described above, have several shortcomings. For example, many cooker/smoker systems produce an insufficient amount of smoke to provide adequate flavoring of food within the cooking environment. In cooker/smoker systems, heat from the heat source and oxygen are required for smoke-releasing material to ignite, smolder and produce smoke. The heated air from the heat source heats the smoke-releasing material and provides the oxygen necessary for igniting and smoldering the smoke-releasing material. However, because the oxygen level of the heated air has

been depleted due to the combustion of the heat source material, such as natural gas or propane, there can be insufficient oxygen present in the heated air to provide sufficient ignition and/or smoldering of the smoke-releasing material. Accordingly, an insufficient amount of smoke for flavoring the food can result.

**[0007]** Further, conventional smoker/cooker systems typically do not provide an easy and sufficient way to adjustably control the amount of smoke being produced. For example, if a greater or lesser amount of smoke is desired, many conventional smoker/cooker systems require an alteration of the heat being produced by the heat source or removal of the smoke-generating device from the food cooking environment. Increasing the heat may cook the food at an undesirable rate. Further, removing the smoke-generating device during a cooking operation can be cumbersome and dangerous. Many conventional smoker/cooker systems do not provide a way to easily adjust the smoke level during a cooking operation independent of the heat source.

### BRIEF DESCRIPTION OF THE DRAWINGS

**[0008]** FIG. 1 is a perspective view of a smoker device according to one exemplary embodiment.

**[0009]** FIG. 2 is a perspective view of a smoke-releasing material holder and diffusion portion of the smoker device of FIG. 1.

**[0010]** FIG. 3 is a top plan view of the smoke-releasing material holder and diffusion portion of the smoker device of FIG. 1.

**[0011]** FIG. 4 is an elevation end view of the smoke-releasing material holder of the smoker device of FIG. 1.

**[0012]** FIG. 5 is an elevation side view of the smoke-releasing material holder of the smoker device of FIG. 1.

**[0013]** FIG. 6 is a cross-sectional front view of a grill having a smoker device according to an exemplary embodiment.

### DETAILED DESCRIPTION

**[0014]** Described herein are several embodiments of a smoker device for use with a cooking apparatus. In certain implementations, the smoker device is positioned within the food cooking environment of the cooking apparatus to provide an economical and convenient way to smoke food. The smoker device, according to some implementations, provides increased smoke-producing characteristics compared to conventional smoker devices. Further, in some embodiments, the smoker device facilitates adjustable, in situ, controllability of the smoke level produced by the device independent of the level of heat produced by the heat source of the cooking apparatus.

**[0015]** Referring to FIG. 1, and according to one embodiment, a smoker device 10 includes a smoke-releasing material holder 20, e.g., a container, receptacle or box, and an air induction system 30 coupled to the holder. With reference to FIG. 2, the material holder 20 includes a closed bottom end 22 and an open top end 24. The closed bottom end 22 is coupled to the open top end 24 by sides 26 extending between, and generally transversely relative to, the top and bottom ends. The sides 26 include first spaced-apart sides 26A extending parallel to each other and second spaced-apart sides 26B extending parallel to each other and transversely from the sides 26A. As shown, the bottom end 22 and each side 26 has a generally rectangular shape, such that

the holder **20** has a generally elongate cube shape. However, in some embodiments, the holder **20** can have any one of various shapes, such as cylindrical, prismatic, cubic, etc. In the illustrated embodiments, the bottom end **22** and sides **26** are made from substantially planar, thin-walled sheets of a conductive material. Preferably, the conductive material is a rigid, strong, and durable material with a high melting temperature, such as steel.

[0016] Also, although in the illustrated embodiments, the closed end **22** of the holder **20** is completely closed, or free of openings, and the open end **24** of the holder **20** is completely open, or free of obstructions, in other embodiments, the closed end can be partially closed and the open end can be partially open. For example, in certain implementations, the closed end **22** can have one or more openings (not shown) large enough to allow direct heat transfer from a heat source to the smoke-releasing material, but small enough to prevent the smoke-releasing material to fall through. Likewise, in certain implementations, the open end **24** can be a partially open cover, such as a mesh, vented, or louvered cover, to control or direct smoke output, or keep contaminants, such as grease, from accumulating within the holder.

[0017] The holder **20** includes a material containment area **28** defined by the bottom end **22** and sides **26**. The material containment area **28** is sized to receive and contain a predetermined amount of smoke-releasing material. Accordingly, the bottom end **22** and sides **26** can have any of a number of shapes and sizes to define a material containment area **28** sufficiently sized to accommodate a desired amount of smoke-releasing material (not shown) for any of variously sized and shaped cooking apparatus.

[0018] Referring back to FIG. 1, the air induction system **30** includes an air diffusion portion **32**, feed line **34** and forced air actuator **36**. The air diffusion portion **32** is positioned within the material containment area **28** and includes one or more air diffusers **38** coupled to and in air receiving communication with an air manifold **40** (see FIG. 3). Each diffuser **38** includes a length of tubular material defining a generally hollow passageway through which air is allowed to flow. As shown, the diffusers **38** each have a generally square cross-sectional shape. However, in other embodiments, the diffusers **38** have any of various other cross-sectional shapes, such as, for example, rectangular, triangular, etc.

[0019] A plurality of spaced-apart apertures **42** are formed in and extend through the walls of each diffuser **38**. As shown, apertures **42** are formed in each of the sides of the diffusers exposed to the material containment area **28** and evenly distributed along a length of the diffusers **38**. The apertures **42** can be formed in the diffusers **38** in any of various patterns, such as staggered, or be randomly arranged. Similar to the diffusers **38**, the manifold **40** includes a length of tubular material defining a generally hollow passageway. An aperture (not shown) corresponding with each diffuser **38** is formed in the wall of the manifold. Each diffuser **38** is secured to the manifold **40**, such as by welding, fastening or other coupling technique, over a respective aperture in the manifold such that the hollow passageway is in alignment and air receiving communication with a respective aperture of the manifold. Alternatively, the diffusers **38** can be formed integrally with, e.g., as a one-piece unit with, the manifold **40**.

[0020] In the illustrated embodiments, the diffusion portion **32** includes three spaced-apart diffusers **38** extending generally parallel to each other and the first sides **26B** along a length of the material holder **20**. In other embodiments, the diffusion portion **32** can include more or fewer than three diffusers **38** and the diffusers can extend generally parallel to second sides **26A** along a width of the material holder **20**. Alternatively, the diffusers **38** can be other than substantially straight. For example, the diffusers **38** can be generally circular or ovalar, such as with a holder **20** having a circular or ovalar shape.

[0021] Referring to FIG. 1, the air manifold **40** is in air receiving communication with an outlet end **44** of the feed line **34** via a fitting **46** formed in or coupled to a side **26**, such as side **26A**, of the holder **20**. The feed line **34** includes a length of tubing, piping or braided line made of a heat resistant material, such as steel, fiberglass or copper, or coated with a heat resistant material, such as silicone rubber or a ceramic material.

[0022] The inlet end **48** of the feed line **34** is coupled to and in air receiving communication with the forced air actuator **36**. The forced air actuator **36** can be any of various devices, such as a pump, fan, bellows, etc., configured to pump air into the feed line **34**. In certain embodiments, the forced air actuator **36** is selectively controllable to force air into the feed line **34** at different rates. In certain implementations, the smoker device **10** includes a forced air actuator control assembly (not shown) in communication with the actuator **36**. The control assembly can be operable by a user of the smoker device **10** to vary the rate of air into the feed line **34**, and thus the material containment area **28** as will be described in more detail below.

[0023] In certain embodiments, the smoker device **10** is used in cooperation with a grill apparatus, such as gas grill **100** shown in FIG. 6. Gas grill **100** can be any of various gas grills known in the art or a gas grill specifically suited for use with the smoker device **10**. In other words, the smoker device **10** can be retrofitted into an existing gas grill **100** to convert the grill from a cooker to a combination smoker/cooker. Alternatively, the smoker device **10** can be installed in a gas grill at the point of manufacture of the grill and sold to an end user as a combination smoker/cooker.

[0024] Referring to FIG. 6, gas grill **100** includes a body **110** having a base **120** and lid **130**. The base **120** defines a generally hollow space between a lower end **122** and an open upper end **124**. Similarly, the lid **130** defines a generally hollow space between an upper end **132** and an open lower end **134**. The lid **130** is coupled to the base **120** and movable (e.g., pivotable, rotatable, slidable, etc.) between an open position and a closed position. When in the closed position (shown in FIG. 6), the interface between the base **120** and lid **130** forms an at least partial seal and the open upper end **124** of the base and open lower end **134** of the lid are adjacent each other such that the hollow spaces of the base and lid form a food cooking/smoking environment, or space, **140**. The grill **100** includes a heat source, such as burners **150**, secured to and positioned within the base **120** intermediate the lower end **122** and open upper end **124**. The grill **100** further includes a food rack, or grid, **160** positioned within the base **120** intermediate the burners **150** and the open upper end **124**.

[0025] The holder **20** is positioned within the base **120** intermediate the burners **150** and the food rack **160**. The holder **20** extends generally parallel to the food rack **160**

with the open top end **24** facing the food rack. In the illustrated embodiments, the holder **20** is supported by a sear plate **154** secured to the base **120** over the burners **150**. In many conventional grills, one or more sear plates are provided to protect the burners from falling grease and food. Further, sear plates can conduct heat from the heat source and provide radiant heat to the food rack **160**. Similarly, in addition to, or in place of, sear plates, some conventional grills include a briquette rack and briquettes positioned between the heat source and the food rack. In these grills, the holder **20** can be supported on top of the briquettes. Alternatively, the holder **20** can be mounted to the base **120** by one or more stand-off brackets (not shown) secured to the base.

[0026] In certain implementations, the holder **20** is removable if, for example, the smoke-releasing material needs to be replenished, the holder requires maintenance or replacement, or the user no longer desires to smoke food and wants to use the grill only to cook food. Accordingly, in certain implementations, the grill **100** can operate in a smoker/cooker mode by positioning the holder **20** within the grill and operating the smoker device **10** as described above, or in a cooker mode by removing the holder from the grill.

[0027] The feed line **34** extends from the holder **20** at a location within the base **120** to the forced air actuator **36** at a location external to the base via an aperture **126** formed in the base. As shown, the feed line **34** extends through a small aperture formed in a side of the base. However, in certain implementations, the grill can include a slot formed in the base through which the holder **20** is inserted to position the holder within the grill and from which the holder is removed. In these implementations, the feed line **34** can extend through the slot when the holder is positioned within the grill.

[0028] In use, the lid **130** is moved into the open position and food to be cooked is placed on the rack **160**. Prior to, or after, placing food on the rack **160**, one or more of the burners **150** are activated to produce a flame **152**. Heated gas from the flame rises and cooks the food. As the heated gas rises, it contacts the holder **20** causing it to rise in temperature. Once the temperature of the holder **20** reaches a predetermined level, which is dependent on the combustibility of the smoke-releasing material contained by the holder, the material ignites, begins to smolder, and produces smoke. The smoke is released into the food cooking environment **140** via the open top end **24** of the holder **20**. As discussed above, the heated gas rising into contact with and around the holder **20** has a substantially reduced level of oxygen in relation to the air outside of the grill due to the combustion the heat source fuel.

[0029] The forced air actuator **36** is operable to supply fresh air from outside the grill **100** into the material containment area **28** of the holder **20** via the air diffusion portion **32**. More specifically, the forced air actuator **36** generates air flow through the air feed line **34** and into the air manifold **40**. From the air manifold **40**, the air flows into the air diffusers **38** at an approximately equivalent rate and through the

apertures **42** into the material containment area **28**. In certain embodiments, the apertures **42** are arranged to evenly distribute fresh air into the material containment area **28**. The fresh air entering the material containment area **28** from outside the grill **100** contains more oxygen than the heated gas within the grill **100**. The additional oxygen in the material containment area **28** improves ignition and smoldering performance of the smoke-releasing material to enhance the smoke output of the material.

[0030] In certain embodiments, the smoke output of the smoker **10** can be selectively controlled by operation of the forced air actuator **36** to adjust the rate of air flow into the material containment area **28**. If a user desires more smoke flavoring in the food being cooked, the air flow rate into the material containment area **28** can be increased by increasing the forced air output of the actuator **36**. Likewise, if a user desires less smoke flavoring in the food being cooked, the air flow rate into the material containment area **28** can be decreased by decreasing the forced air output of the actuator **36**. Unlike conventional smoker/cooker devices, a user need not adjust the heat output of the heat source, or add smoke-releasing material to or take smoke-releasing material away from the holder, to adjust the amount of smoke in the grill. Further, operation of the forced air actuator **36** can be initiated by controls, such as control knob **170**, in communication with the actuator and positioned near the burner control knobs, such as knob **180**. Accordingly, the cooking operation of the grill **100** and smoking operation of the smoker **10** can be controlled from one convenient location.

[0031] In view of the many possible embodiments to which the principles of the disclosed smoker may be applied, it should be recognized that the above embodiments are only examples of the smoker and should not be taken as limiting the scope of the invention.

What is claimed is:

1. A method of supplying smoke within a grill, comprising:
  - positioning a smoke-releasing material container within a grill between a heat source and food grid of the grill;
  - supplying air from a location external to the grill to the smoke-releasing material container; and
  - heating smoke-releasing material contained within the container.
2. The method of claim 1, wherein supplying air comprises actuating a forced air device in air transmitting communication with the smoke-releasing material and positioned external to the grill.
3. The method of embodiment 1, further comprising smoldering smoke-releasing material contained within the smoke-releasing container.
4. The method of embodiment 1, wherein supplying air comprises transmitting air through an air manifold system positioned within the smoke-releasing container.
5. The method of embodiment 4, wherein transmitting air comprises transmitting air through a plurality of spaced-apart air diffusers each having a plurality of air outlets.

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